

B.TECH CSE SEM IV		
	GOOGLE CLOUD	3 CREDITS

Learning Objectives:

- 1 Discuss what the cloud is and why it's a technology and business game changer.
- 2 Describe the different ways a user can interact with Google Cloud Platform.
- 3 Discover the compute options in Google Cloud Platform.
- 4 Implement a variety of structured and unstructured storage models.
- 5 To discover a variety of managed big data services in the cloud

GCCF SYLLABUS

UNIT 1	Introduction, Cloud Computing, Cloud Vs Traditional Architecture, IAAS, PAAS, SAAS, Google Cloud Architecture, Google Cloud Console, Projects, Billing in Google Cloud, Install and Configure SDK, Use Cloud Shell, Google Cloud APIs, Cloud Console Mobile App
UNIT 2	Compute Options in The Cloud, Exploring IAAS With Compute Engine, Configuring Elastic Apps with Autoscaling, Exploring PAAS With App Engine, Event Driven Programs with Cloud Functions, Containerizing and Orchestrating Apps With GKE, Storage Options in The Cloud, Structured and Unstructured Storage in The Cloud, SQL Managed Services, Cloud SQL, Cloud Spanner as A Managed Service, Cloud Datastore a NoSQL Document Store, NoSQL Managed Services, Cloud Bigtable As a NoSQL Option
UNIT 3	The Purpose of APIs, Cloud Endpoints, Using Apigee, Managed Message Services, Cloud PUB/SUB, Security in The Cloud, Shared Security Model, Encryption Options, Authentication and Authorization with Cloud IAM, Best Practices for Authorization Using Cloud IAM
UNIT 4	Introduction to Networking in The Cloud, Defining A Virtual Private Cloud, Public and Private IP Address Basics, Google's Network Architecture, Routes and Firewall Rules in The Cloud, Multiple VPC Networks, Building Hybrid Clouds Using VPNs, Interconnecting and Direct Peering, Load Balancing, Infrastructure as Code IaC, Cloud Deployment Manager, Monitoring and Managing Your Services, Applications and Infrastructure
UNIT 5	Big Data Managed Services in The Cloud, Bigdata Operations with Cloud Dataproc, Bigquery, Google's Enterprise Data Warehouse, Machine Learning, Machine Learning and Google Cloud Qwik Draw, Qwik Draw Screencast, Building Bespoke Machine Learning Models with AI Platform, Cloud AutoML, Google's Pre-Trained Machine Learning APIs

Course Outcomes	Program Outcomes												Program Specific Outcomes		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1		M							M		H		M	H	
CO2	L		M		H		L	L		M		M		H	M
CO3		M		H	M	L		L			H		M	H	
CO4	L		H	M			L	M	M		H		M	H	

CO5			H	M	H	L				H		M	H	H	
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